Venkata Lakshmi Parimala Pasupuleti

LinkedIn | GitHub | Portfolio | Virginia, USA

PROFESSIONAL SUMMARY

Software Engineer with 8+ years of production engineering and agentic AI systems. Built LeaseI -- an 8-agent LangGraph human-in-the-loop (HITL) pipeline with conditional escalation, 93.6% precision, 87.9% F1, and 100% citation correctness across 1,240 lease PDFs using Retrieval-Augmented Generation (RAG) (github.com/venkatalakshmiparimala/Lease_complaince_rag). Shipped distributed backend services at Entrata for 100K+ monthly active users across 120+ Agile sprints at 99.5% uptime, including Entrata's first production AI pipeline across 1.6M real financial records (Python, PyTorch, OpenAI API). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Multi-Agent Orchestration (LangGraph) | RAG Pipelines & Vector Databases | HITL Workflow Design | MLOps & LLM Evaluation | Distributed Systems & Backend APIs | Python · TypeScript · FastAPI | CI/CD · Docker · AWS | Agentic AI & LLM Integration

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern (Unpaid)

- Built an agentic mission-planning system as a LangGraph stateful graph (parse - plan - retrieve - validate - conditional routing) delivering risk-aware compliance strategy: compliant missions route to export, non-compliant ones route to rejection with grounded reasons.
- Implemented a RAG layer (retrieval over FAA Part 107 regulatory corpus) with cited sources -- every compliance check functions as a hypothesis test grounded in source regulation; designed a backend-agnostic retriever with a documented ChromaDB/embedding upgrade path.
- Engineered validation guardrails with conditional routing enforcing FAA altitude limits, battery budgets, and geometric no-fly-zone checks -- blocking non-compliant missions before export.

George Mason University -- College of Engineering and Computing

Aug 2025 - May 2026

Graduate Teaching Assistant (Part-time, On-site)

- Mentored 600+ students across multiple sections of MIS 303 in Python programming (OOP, automation scripting, data structures) and MS Access database design (ER modeling, SQL, normalization); provided one-on-one debugging support and lab guidance.

George Mason University

Aug 2024 - Aug 2025

Research Associate

- Achieved R2 = 0.977 regression accuracy across 190+ countries by building multi-source prediction models (Python, R) on WHO and UK Road Safety datasets -- data collection, transformation, hypothesis testing, and validation pipelines.
- Surfaced anomalies manual review missed across 3 research pipelines by applying Isolation Forest to automate data integrity checks -- improved downstream model reliability.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 - Aug 2024

Senior Software Engineer (Remote)

- Shipped 20+ full-stack production modules (PHP, Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users -- 99.5% uptime over 6+ consecutive months; led sprint planning using Planning Poker.
- **Shipped Entrata's first production AI pipeline** across 1.6M real financial records (Python, PyTorch, OpenAI API) -- automated data processing workflows using custom neural network architecture; output validation, quality gates, and compliance controls at launch.
- **Led microservices architecture redesign** using Docker -- 50% improvement in scalability, 30% reduction in deployment time; converted 20% of product modules into service-based architecture, reducing interdependencies.
- Built event-driven data pipelines using RabbitMQ for asynchronous processing across distributed financial and compliance services.
- **Achieved 100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines (encrypted 100% of resident data); zero security incidents.
- Optimized 200+ SQL queries across all client environments (30% performance gain) through execution plan tuning, indexing, partitioning, and caching; eliminated 80% of manual operational workload via Python/Bash automation; mentored 5+ junior engineers.

Entrata India, Pune

Jul 2017 - Sep 2018

Associate Software Engineer

- Developed full-stack property management applications (PHP, Python, PostgreSQL, JavaScript); migrated thousands of client documents to AWS S3, resolving 90% of legacy space constraints; redesigned document storage database architecture for hundreds of thousands of records.

PROJECTS

Leasel: AI Document Intelligence & HITL Support Investigation Platform 2026

Python, LangGraph (8-agent), FastAPI, TF-IDF, OCR, SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an 8-agent LangGraph human-in-the-loop (HITL) pipeline (Triage - Retrieval - Configuration - Historical - Reasoning - Validation - Human Escalation) with conditional escalation edge when confidence is too low -- agent memory systems track state across steps; LLM reasons, human/deterministic logic decides.
- Built a Document Intelligence mode: OCR-fallback ingestion, section-aware chunking, TF-IDF RAG retrieval, confidence-scored answers with 100% citation correctness, and human-review gate for regulated decisions.
- Achieved 93.6% precision and 87.9% F1 across 10 compliance rules on 1,240 residential lease PDFs; pass@1 81.7% via pass@k harness with multi-trial sampling, regression gate, and LLM-as-judge scoring.

AI on FHIR: NLP-Driven Medical Records Query System 2024

Flask, LLMs, FHIR, React, Docker, spaCy, BERT, Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced manual clinical retrieval time ~60% and enforced HIPAA-compliant RBAC across structured medical records by building a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and Retrieval-Augmented Generation -- production LLM integration into enterprise developer tools for clinical teams.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

Languages: Python, TypeScript, JavaScript, Node.js, SQL, PHP, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Agentic AI, Multi-Agent Orchestration, HITL, LLM Orchestration, RAG Pipelines, ChromaDB, Vector Databases, Semantic Search, Embeddings, LLM Evaluation, MLOps, LLM-as-Judge, Agent Observability, Agent Tracing, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, FastAPI, REST APIs

ML & Data: PyTorch, TensorFlow, XGBoost, scikit-learn, Isolation Forest, Anomaly Detection, Feature Engineering, NLP, BERT, spaCy, Hugging Face, Pandas, NumPy, Tableau, Power BI

Cloud & Infra: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), Microservices, Event-Driven (RabbitMQ), PostgreSQL, MySQL, Oracle, Git, Linux, Agile/Scrum